

**Domain:** Road signs as used in the European Union.

**Purpose:** To identify sign types based on properties and determine which agents (like autonomous vehicles) they apply to.

### Competency Questions:

#### Taxonomy & Hierarchy (Structure)

1. What are the primary subclasses of the **RoadSign** class defined in the ontology?
2. Is a **ProhibitorySign** logically classified as a subclass of **RegulatorySign**?
3. Are the **DangerWarningSign** and **InformativeSign** classes mutually disjoint?
4. What is the direct parent class of the **MandatorySign** category?
5. Does the ontology successfully classify individual **Sign\_A1a** as a **DangerWarningSign**?

#### Visual Attributes (Shapes & Colors)

6. Which sign individuals possess the property **hasShape** linked specifically to the value **shape\_triangle**?
7. What is the universally required shape instance for all **MandatorySign** instances currently in the dataset?
8. Which specific signs share the property **hasGroundColour** linked to **color\_blue**?
9. What is the required **hasBorderColour** for a standard prohibitory sign, such as **Sign\_C2**?
10. What unique shape instance is assigned to **Sign\_B2a** (the standard STOP sign)?

#### Symbols & Meanings (Pragmatics)

11. Which exact symbol instance is used to warn drivers of a steep descent on **Sign\_A2a**?

12. What specific sign ID is used to represent the "Compulsory Roundabout" regulation?
13. Which priority sign is linked to the **symbol\_end\_of\_priority** instance?
14. What symbol differentiates **Sign\_C11a** (No Left Turn) from **Sign\_C11b** (No Right Turn)?
15. Which sign individuals in the ontology are linked to **symbol\_pedestrians** or **symbol\_pedestrian\_crossing**?

### Complex Logic & Cross-Referencing (Dataset Queries)

16. How many total **PrioritySign** instances are explicitly defined and searchable in the ontology?
17. Which individual signs combine both **shape\_circle** and a **hasBorderColour** of **color\_red**?
18. Which **MandatorySign** instances dictate a compulsory path for specific road users (e.g., pedestrians, cyclists, equestrians)?
19. Are there any **DangerWarningSign** instances that feature an additional **color\_black** border alongside the standard red one? (*Answer: Yes, the St. Andrew's Cross signs A28a/c*).
20. Which specific signs in the **ProhibitorySign** class are used to indicate speed limits or the end of speed limits?

**LLM: Gemini**

**LLM Prompt:**

Role: You are an expert Ontology Engineer specializing in traffic regulations and knowledge representation.

Objective: Generate OWL2 individuals (instances) for all \_\_\_\_\_ Signs described in Section \_ of the Vienna Convention, and link them to their respective visual attributes using object property assertions.

Scope & Domain:

Domain: Section \_ of Part I, Annex 1 of the Vienna Convention on Road Signs and Signals.

Key Entities: Create individuals for every sign model in this section. You must assert that each is an instance of the \_\_\_\_\_ Sign class.

Properties: For every sign, generate object property assertions using the properties: hasShape, hasGroundColour, hasBorderColour, and hasSymbol.

Value Instances: Assume the basic color and shape instances already exist (e.g., color\_red, color\_blue, shape\_circle). Create new instances for specific symbols as needed, enforcing lowercase with underscores (e.g., symbol\_no\_entry).

Boundaries: Do NOT redefine the classes or the object properties themselves. Only output the XML code for the owl:NamedIndividual declarations and their property assertions.

Structural Requirements:

Use standard RDF/XML syntax.

Ensure all URIs align with the generic base URI:

<http://www.ontology.org/traffic/vienna-convention#>.

Enforce a strict naming convention for sign IDs: Sign\_ followed by the exact alphanumeric code (e.g., Sign\_C1a).

Output Format: Provide ONLY the raw RDF/XML code snippet containing the individuals.